ANDREW STEVENS

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Website Portfolio: www.andrewstevens.engineering

EDUCATION

Virginia Tech– Blacksburg, VA **Expected: May 2029**
B.S. Electrical Engineering - Junior standing by credits **GPA: 4.0**
Selected Courses: Linear Algebra, Digital Systems, Embedded Systems, Electromagnetic Physics, Differential Equations

PROFESSIONAL EXPERIENCE

Internship - Software Engineer **May 2026 - August 2026**
Hitachi Energy – Raleigh, NC
- Scoped and planned an autonomous Python report generator, defining requirements and milestones
- Built the application to parse transformer service test files and generate formatted reports, reducing documentation time from 4–6 hours to approximately 5 minutes (~98% reduction)

Current Owner And Founder **March 2021 - Present**
Andy's 3D Workshop – Ashburn, VA
- Established the company and developed business plans for optimized sales growth
- Designed and prototyped all products, including 1 patented with US Patent & Trademark Office
- Sold products to businesses and partnered with 2 local stores in Northern Virginia
- Created the company's online store on Etsy with over 1,600 sales and a 5-star rating

ADDITIONAL EXPERIENCE

Connected & Autonomous Vehicles (CAV) Team Member **January 2026 - Present**
Hybrid Electric Vehicle Team – Virginia Tech
- Designed Cooperative Adaptive Cruise Control in simulink to increase energy efficiency of Cruise Control by 60.3%
- Improved existing ROS2 in Python Cruise Control to account for the changed Cooperative Adaptive Cruise Control
- Training in high-voltage safety systems, powertrain architecture, vehicle sensing, control integration, and cross functional engineering processes for an electric vehicle that is part of the EcoCar Competition
- Designed and implemented intersection crossing logic in Simulink and control blocks to govern vehicle decision-making

ECE Subteam **September 2025 - Present**
VT CRO Workcell – Virginia Tech
- Designed calibration and position script in Python for the system gantry that achieved precision of ± 0.05mm
- Designed custom PCBs for daisy chaining limit switches for plate storage detection, eliminating 59 loose wires
- Designing a printer UI and control system to interface with the printers, providing status feedback and parameter control

ECE Lead **August 2026 - Present**
VT CRO Workcell – Virginia Tech
- Lead electrical and computer engineering architecture for an autonomous greenhouse robotic system, defining system structure, interfaces, and technical direction
- Recruited and interviewed prospective engineers, onboarded new members, and communicated project goals, system architecture, and technical requirements

Testing & Electronics **September 2025 - March 2026**
Baja SAE – Virginia Tech
- Developed a custom microcontroller GPS tracker to track position and velocity, to improve race performance analysis
- Made a Python interface to visualize sensor data at any point during a race, improving diagnostics and evaluation
- Developing telemetry for real time monitoring of vehicle health in competitions, improving reliability and safety
- Designed custom professional PCBs in KiCAD, improving system reliability for the Baja electronics suite

SKILLS

- Technical Skills: PCB Design, Embedded Systems, Digital Systems, Circuit Design, C++, Python, C#, MATLAB, Simulink, ROS2, Git, Microcontrollers, OpenCV